ARGUS TECHNOLOGIES • ENTERPRISE CNC SOFTWARE

Metal Calculator & Quotation Suite

Complete System User Manual, Technical Specifications & Visual Feature Guide

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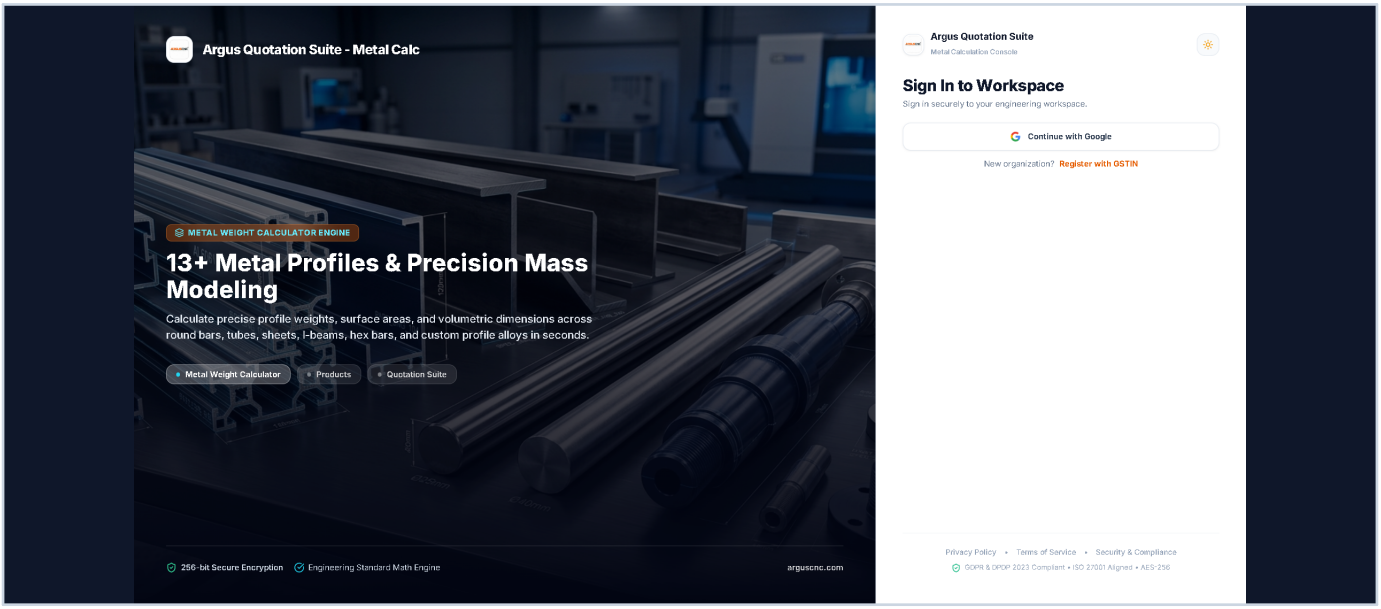
Official Documentation & Operations Guide
System Release: Version 7.5.0 (Full-Stack Enterprise Edition)
Deployed Organisation: **Argus Technologies** (GSTIN: 33CZEPS8675J1ZN)
Scope: Comprehensive feature-by-feature visual walkthrough with high-resolution system screenshots.

Executive Summary:
The **Argus Metal Calculator & Quotation Suite** is a high-precision, production-grade estimation and commercial quotation platform designed specifically for CNC machine shops, precision fabricators, toolmakers, and metal component manufacturers. It unifies metallurgical weight calculations with an interactive Bill of Materials (BOM) engine, AI-powered 2D blueprint cycle time estimation, live commodity market prices, granular role-based user controls, multi-entity branding, and GST-compliant quotation document generation.

Chapter / Module	Primary Functional Scope	Visual Asset Reference
1. Multi-Tenant Authentication	GSTIN auto-fetch, employee & org login, Google SSO	Login & Gateway Screen
2. Metal Calculator & Vector CAD	8 geometric profiles, metallurgical densities, live SVG	Metal Calculator Tab
3. Commercial Quotation Maker	Multi-item BOM, routing, margins, GST tax calculation	Quotation Maker (Commercial)
4. Technical Workings Sheet	Line-item cycle times, hourly machine rates, labor	Workings Engineering View
5. AI Drawing Estimator	Gemini AI Vision cycle time estimation from 2D drawings	AI Blueprint Estimator Modal
6. Client Directory & CRM	Central client directory, GSTIN parsing, dual views	Clients Library (Grid & List)
7. Products & Assemblies Master	Pre-saved BOMs & routings for repeatable component quoting	Products Master Catalog
8. Terms & Conditions Master	Standard payment, delivery, warranty & escalation clauses	Terms & Conditions Library
9. Quotation History & Dispatch	Centralized audit archive, quote summary, Resend emails	Quotation History Directory
10. Team & Role-Based Access	Employee management, access codes, module rights	Users & Access Directory
11. Organisation Settings	Legal entity data, logo & digital signature, sister concerns	Enterprise Settings Panel
12. Professional PDF Output	Multiple corporate themes, live preview, GST invoices	PDF Theme & Preview Modals

MODULE 01: SECURE ACCESS GATEWAY

1. Multi-Tenant Enterprise Authentication & Onboarding



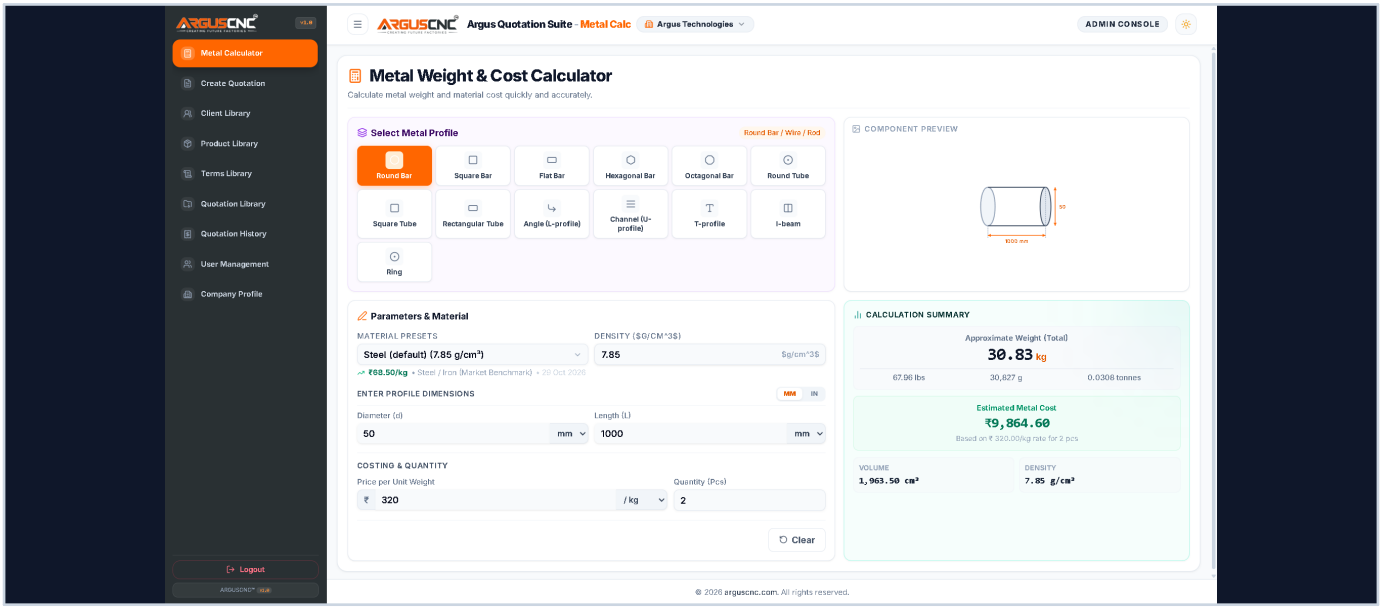
The authentication gateway provides a secure, multi-tenant portal with three distinct access tiers: **Organisation Administrator**, **Standard Employee User**, and **Super Administrator**. It features seamless integration with Indian GSTIN public registries for instant one-click enterprise onboarding.

Authentication Modes	Dual Sign-in: GSTIN / Username & Password, plus One-Click Google Workspace OAuth2 (GIS).
Instant GSTIN Verification	Integrated with Appyflow GST API. Typing a 15-character GSTIN automatically pulls the legal entity name, registered trade name, and tax jurisdiction state.
Role-Based Permissions	Enforces strict RBAC: Standard employees operate under organisation rules, restricting access to sensitive financial settings, master customer lists, or margin figures.
Legal Compliance Hub	Interactive modals for Data Privacy Policy, Terms of Service, and SOC2/ISO-aligned Security & Encryption guarantees.

■ **EFFICIENCY TIP:** Entering an active GSTIN automatically verifies the enterprise against central tax databases, establishing verified company records in less than 2 seconds.

MODULE 02: CORE ENGINEERING CALCULATION

2. Metal Weight Calculator & Vector CAD Blueprint Engine



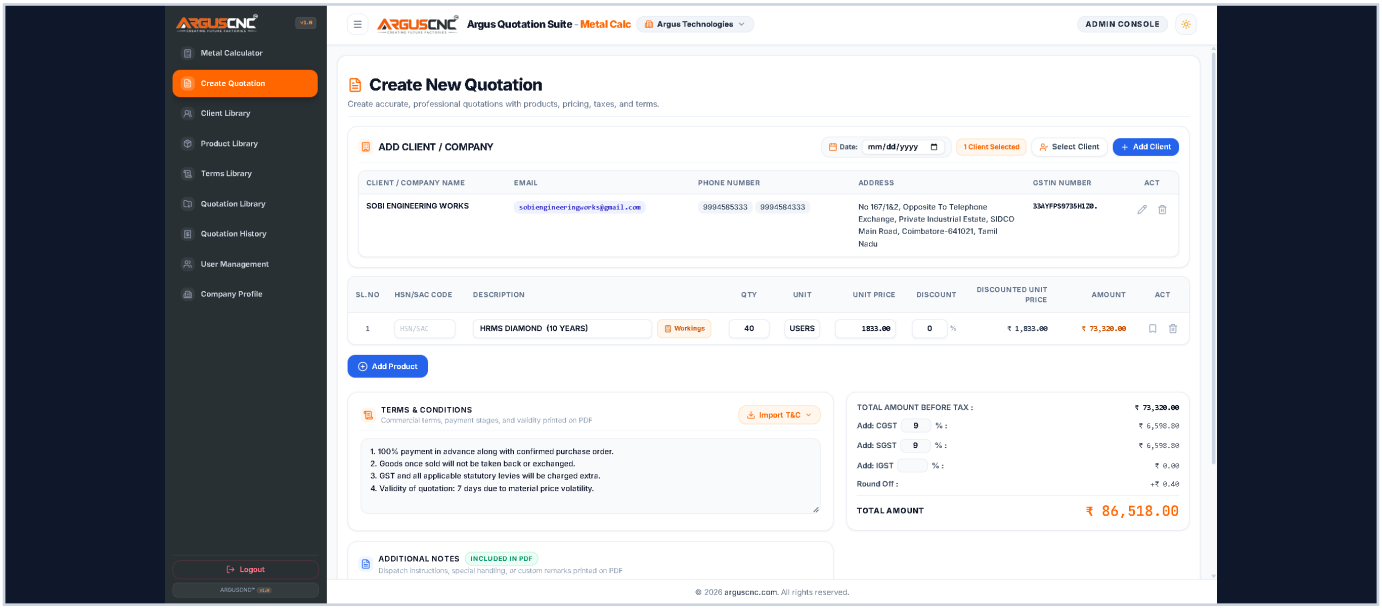
The Metal Calculator is the engineering backbone of the application. It computes high-precision volumetric weight, gross billet mass, scrap recovery salvage value, and raw material procurement costs across 8 geometric profiles. A parametric vector SVG blueprint dynamically updates in real time to visualize input dimensions.

Supported Shapes	Round Bar / Wire, Square Bar, Hexagonal Bar, Flat Bar / Plate, Hollow Pipe / Tube, Square Hollow Section, Circular Ring, Sheet Metal.
Metallurgical Density Library	Pre-calibrated densities: Carbon Steel / Mild Steel (7.85 g/cm³), Stainless Steel 304/316 (8.00 g/cm³), Aluminium Alloys (2.70 g/cm³), Brass (8.50 g/cm³), Copper (8.96 g/cm³), Tool Steels.
Dynamic SVG Blueprint	Parametric vector blueprint renders technical dimension callout lines matching operator input parameters (diameter, width, length, wall thickness).
Scrap Value Recovery	Calculates net component weight vs gross billet raw weight, applying swarf/scrap recovery percentages to offset raw material cost.
Live Metals-API Ticker	Real-time daily metal market commodity pricing ticker ensures quotations reflect current volatile raw material market rates.

■ **WORKFLOW TIP:** Clicking 'Add to Quotation BOM' instantly transfers the calculated piece weight, raw material rate, and dimensions into the active commercial quote with zero manual re-typing.

MODULE 03: COMMERCIAL ESTIMATION ENGINE

3. Commercial Quotation Maker & Costing Suite



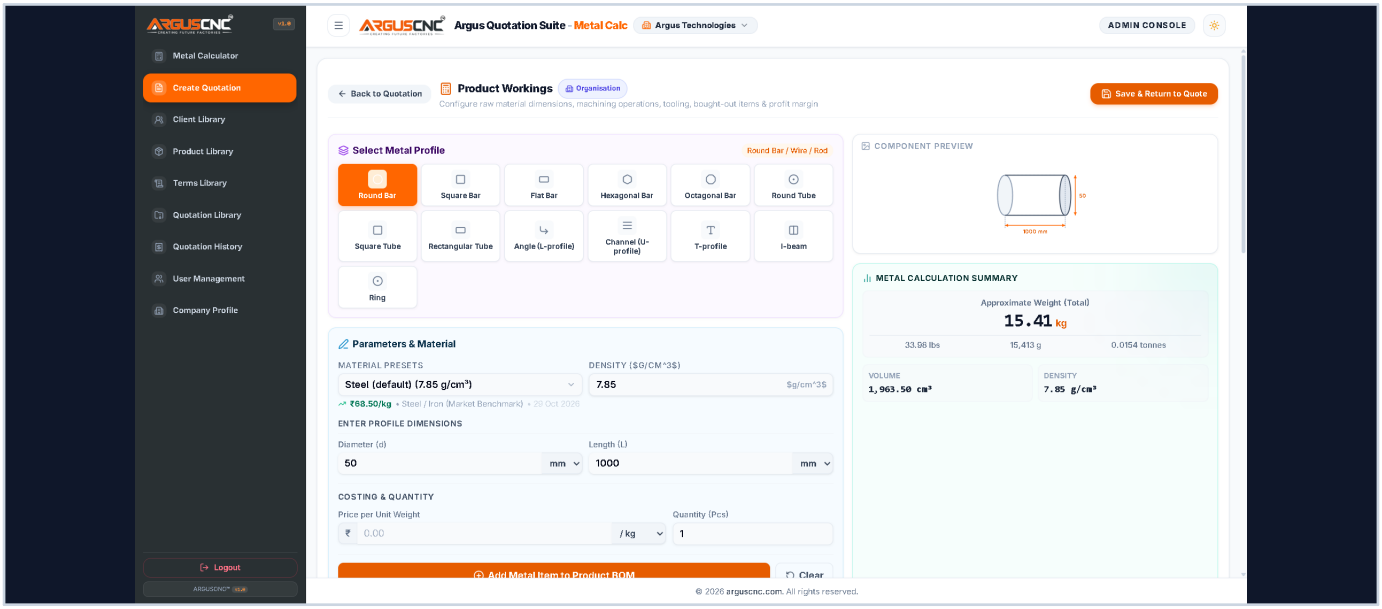
The Commercial Quotation Maker unifies client billing particulars, multi-item Bill of Materials (BOM), comprehensive machining process routings, bought-out accessories, and dynamic profit markups into a unified, statutory GST-compliant quotation builder.

Smart Client Header	Select saved clients from the database. System automatically verifies interstate vs intrastate GST (CGST 9% + SGST 9% vs IGST 18%) based on recipient state code.
Interactive BOM Ledger	Multi-item raw material schedule tracking shape, size, quantity, unit piece weight, raw material cost/kg, and extended raw material subtotal.
Machining Routing Engine	Operations schedule covering VMC 3/4/5-Axis, CNC Turning, Wire EDM, Surface & Cylindrical Grinding, Drilling, Laser Cutting, Heat Treatment, and Surface Finishing.
Dynamic Margin Slider	Interactive profit margin slider dynamically calculates markup value, unit cost vs selling price, gross profit margin %, subtotal, and tax totals in real time.
Terms & Conditions Injector	One-click injection of standardized commercial terms (Advance, Net 30, Validity, Delivery Lead Time) and custom project remarks.

■ **STATUTORY TIP:** The system auto-formats quotation numbers (e.g. MS-Q-1793...) and prevents commercial leakage by ensuring tax rates match state jurisdictions exactly.

MODULE 04: MANUFACTURING COST ANALYSIS

4. Technical Workings Sheet (Deep Engineering Breakdown)



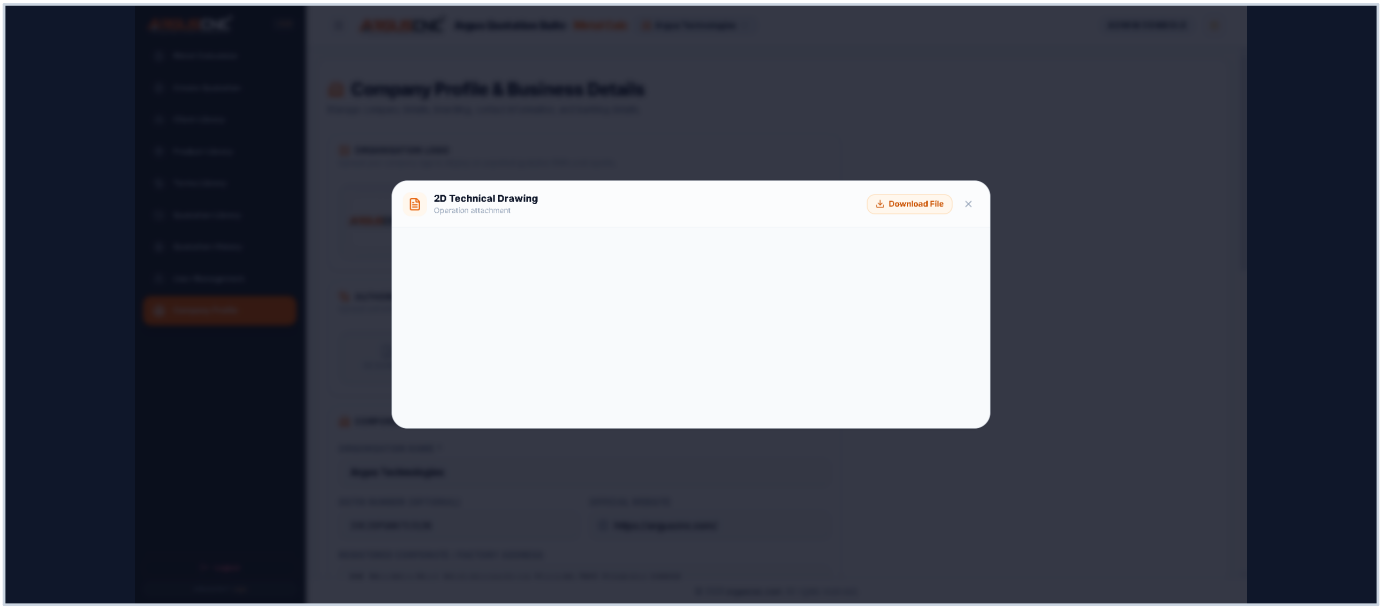
The Technical Workings Sheet provides an engineering-level breakdown of the active product or assembly. It separates raw manufacturing expenses from commercial margins, enabling estimation engineers, workshop supervisors, and managing directors to audit cycle times, setup overheads, and machine hourly charges.

Billet & Material Analysis	Calculates gross raw billet weight required per piece, machining swarf generation, and salvage scrap credit.
Machine Hourly Rates	Configures specific hourly billing rates per machining center (e.g. ■650/hr for 3-Axis VMC, ■1200/hr for 5-Axis, ■450/hr for CNC Lathe).
Setup & Fixture Overhead	Apportions programming time, jig/fixture fabrication fees, and setup machine downtime across the production batch quantity.
Direct vs Quoted Price	Direct visual comparison between pure manufacturing cost (materials + machine run time + tooling) versus final quoted customer price.

■ **AUDIT TIP:** Use the Workings Sheet during internal engineering reviews to verify tool wear allowances and cycle time estimates before signing off on large production runs.

MODULE 05: COMPUTER VISION MACHINING AI

5. AI-Powered 2D Engineering Drawing Cycle Time Estimator



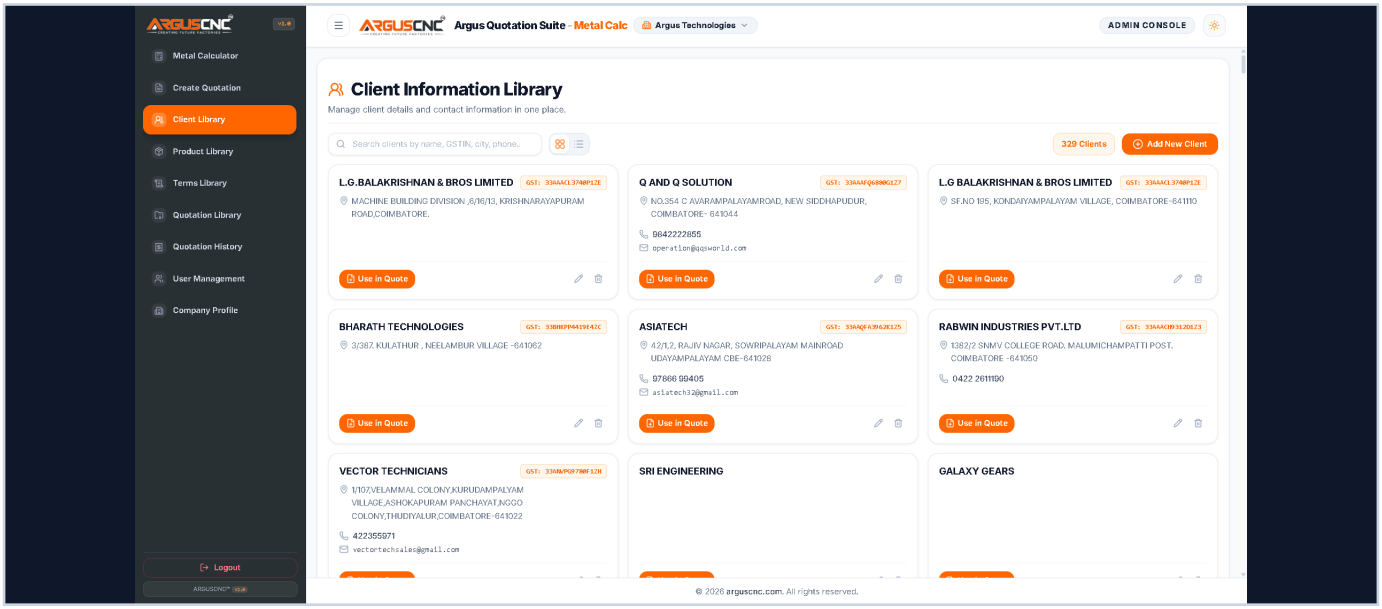
Integrated directly with Google Gemini AI Multimodal Vision, this module eliminates hours of manual drawing analysis. Engineers upload 2D PDF engineering drawings, scanned CAD blueprints, or image files (PNG/JPG). The AI inspects tolerances, dimensions, internal pockets, threads, and surface finishes to predict operational machining cycle times.

Supported Drawing Formats	PDF, PNG, JPG, JPEG, TIFF engineering drawings, CAD exports, and fabrication blueprints.
Feature Inspection Engine	Automatically identifies outer contours, inner bores, tapped holes, counterbores, surface finish Ra requirements, and tight tolerances (H7, g6).
Operational Route Suggestion	Predicts optimal operation sequencing: Facing -> Rough Turning -> Finish Turning -> VMC Pocketing -> Threading -> Grinding.
Cycle Time Estimation	Outputs estimated cycle times in minutes per setup, setup changeover allowance, and recommended cutting parameters (cutting speed Vc, feed f).

■ **PRODUCTIVITY BOOST:** The AI Estimator reduces rapid-fire RFQ turnaround from an average of 45 minutes down to less than 30 seconds per complex engineering component.

MODULE 06: CUSTOMER RELATIONSHIP MANAGEMENT

6. Client Directory & CRM Customer Master



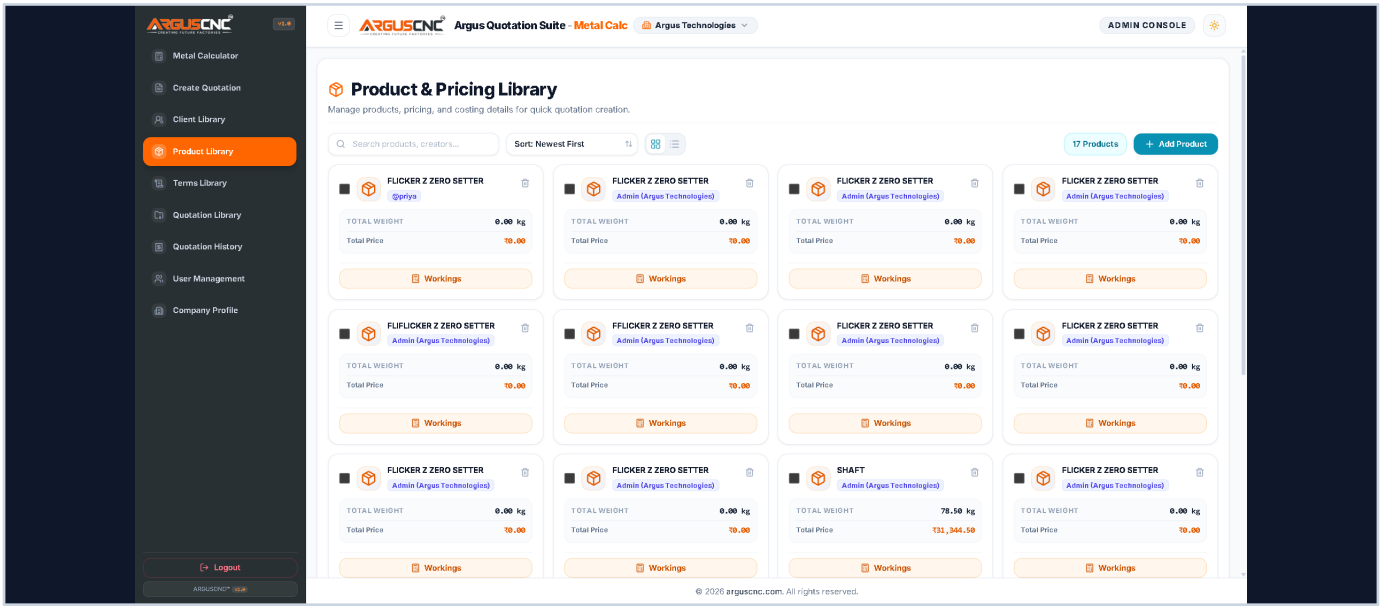
The Client Directory is an enterprise customer repository supporting over 300+ industrial clients. It features dual view modes (Interactive Card Grid vs Compact Sortable Data Table), automated GSTIN validation, multiple contact persons, direct quotation launchers, and full export capabilities.

Dual View Modes	Toggle between Card Grid View (ideal for touchscreens and quick visual browsing) and Compact List Table (optimized for rapid data searching and keyboard navigation).
GSTIN Auto-Fetch & Parsing	Single click pulls legal business names, registered state codes, and verified factory addresses directly from central registries.
Multi-Contact Support	Store multiple purchase managers, accounts contacts, mobile numbers, and direct dispatch emails per client record.
1-Click Quotation Creation	Clicking 'Create Quote' on any client card immediately initializes a new commercial quotation pre-populated with buyer details and appropriate tax schemes.

■ **USABILITY TIP:** Clients are searchable by company name, GSTIN, city, or phone number. Switching between Grid and List view persists across browser sessions.

MODULE 07: STANDARD PARTS & ASSEMBLIES CATALOG

7. Products & Component Assemblies Master



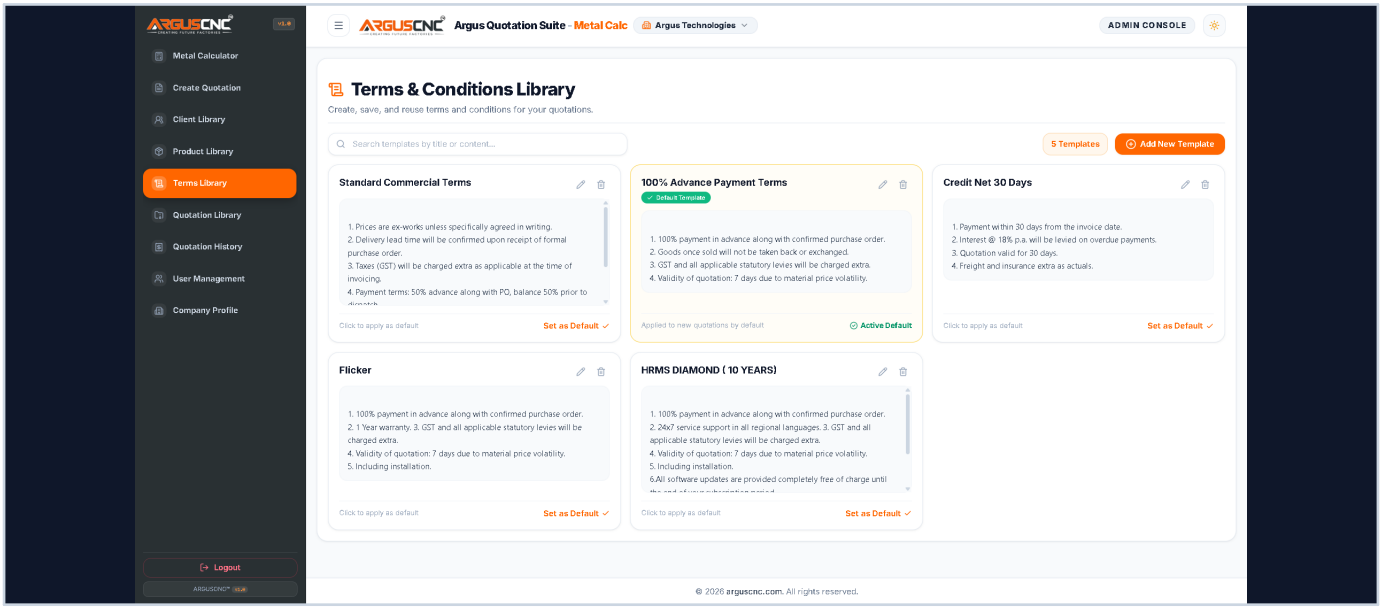
For repeat manufacturing components, custom machinery parts, and proprietary assemblies, the Products Directory stores pre-configured BOMs, standard cutting routes, and unit prices. Assemblies can be added directly into quotations with a single click.

Standard Parts Catalog	Store recurrently machined parts (Shafts, Flanges, Bushes, Spindles, Tooling Blocks) with complete dimensional and material specifications.
Embedded BOM & Operations	Each product retains its complete engineering recipe: raw material billets, VMC machining programs, wire cut timings, and surface finishing.
Instant Quotation Fast-Load	Load high-complexity parts into new customer quotations in 1 second, bypassing manual bill-of-material entry.
Bulk Management & Search	Fast filtering by product name, part code, HSN classification, and category, with bulk deletion and Excel import/export.

■ **BEST PRACTICE:** Save standard parts to the Product Master after finalizing your first prototype run to ensure consistent quotation pricing across your sales team.

MODULE 08: COMMERCIAL TERMS & LEGAL CLAUSES

8. Terms & Conditions Master Template Library



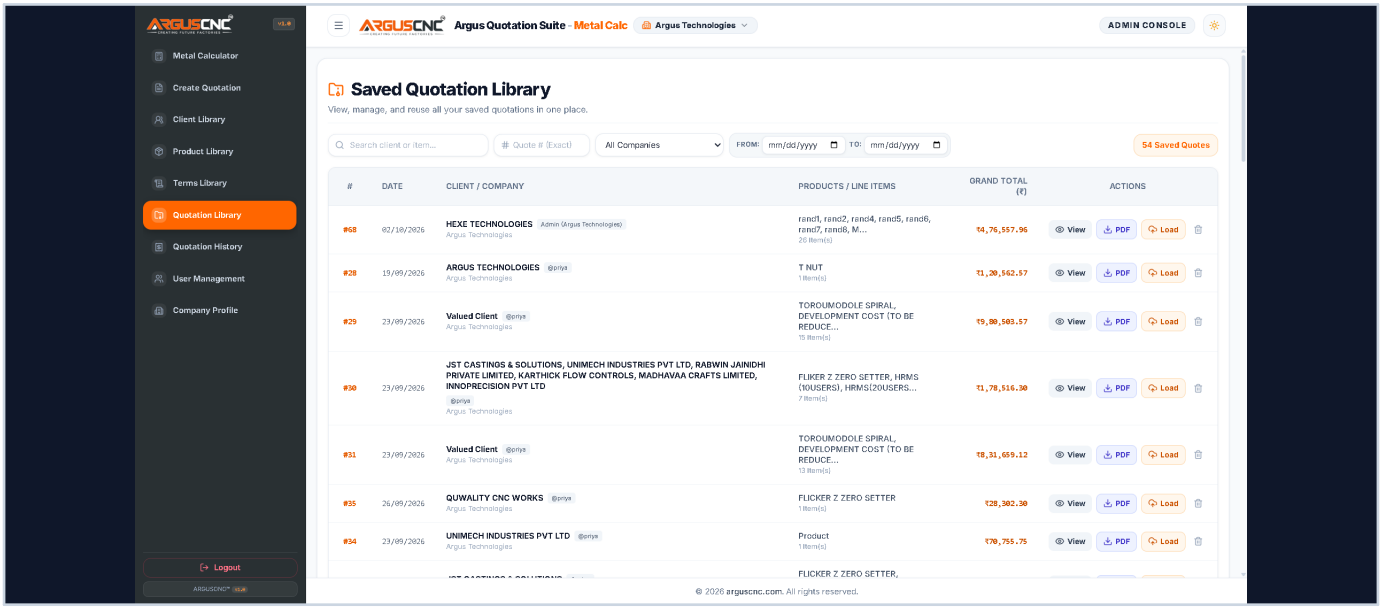
Commercial terms protect manufacturers from raw material price swings, payment defaults, and delivery ambiguities. The Terms Library allows organisations to create, standardize, and maintain multiple commercial clause templates that automatically insert into finalized quotation PDFs.

Pre-Loaded Commercial Templates	Standard Commercial Terms (50% Advance / 50% dispatch), 100% Advance Payment Terms, Credit Net 30/60 Days, Annual Maintenance & Tooling Agreements.
Material Escalation Clauses	Standardized clauses addressing 7-day quotation validity due to volatile domestic and international metal commodity price shifts.
Default Designation	Set a primary default terms template that loads automatically on every new quotation, while allowing quick switching per customer requirement.
Custom Variable Fields	Easily customize delivery lead time (e.g. 2-3 weeks from PO), transit insurance responsibility, and warranty coverage.

■ **RISK MANAGEMENT:** Ensure the 7-day validity clause is active on all raw-material-heavy quotes to protect against London Metal Exchange (LME) and domestic ingot price surges.

MODULE 09: QUOTATION AUDIT & CUSTOMER DELIVERY

9. Quotation History Archive & Email Dispatch



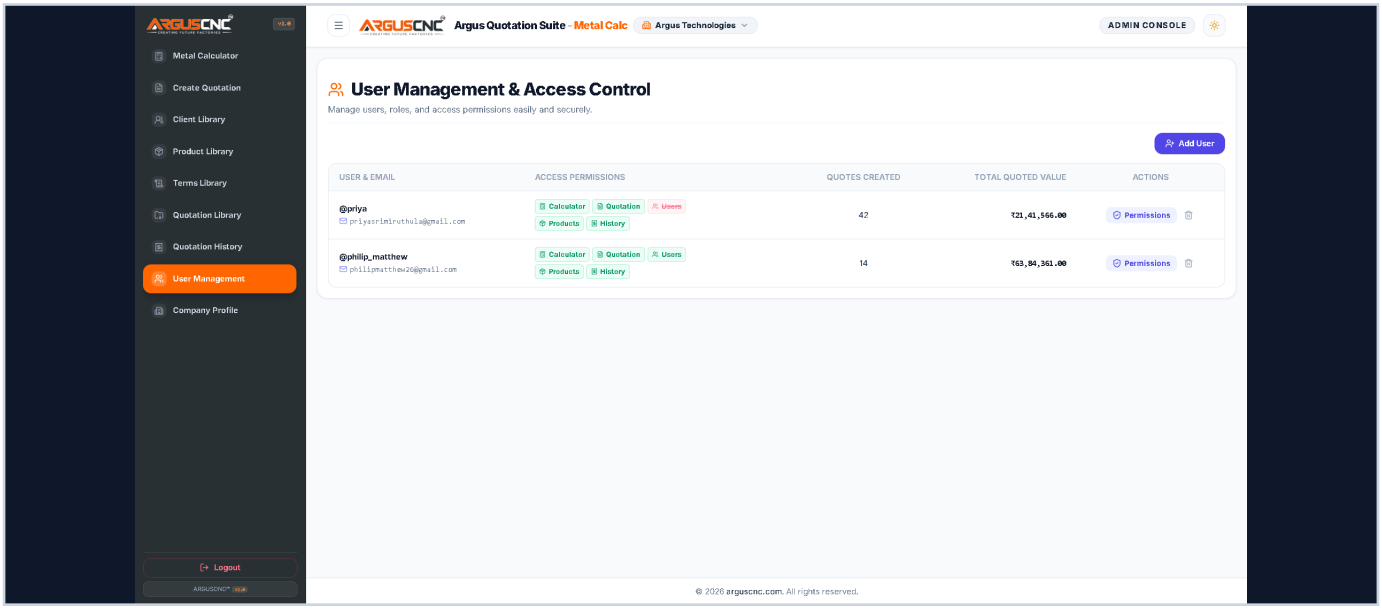
The Quotation History Directory is a centralized, searchable ledger of all issued quotations. Users can inspect line-item breakdowns via the interactive summary modal, regenerate PDFs on demand, and email quotations directly to customers using the integrated Resend API mail server.

Comprehensive Audit Trail	Maintains chronological logs of every generated quotation (e.g. MS-Q-92), creation date, author (@priya, @philip_matthew), customer, and grand total.
Interactive Summary Modal	Clicking any quote opens an instant breakdown modal detailing BOM lines, machining processes, taxes, and customer addresses without opening edit mode.
One-Click PDF Re-Generation	Regenerate pixel-perfect official PDF documents anytime with updated exchange rates or reprint previous revisions.
Integrated Email Dispatch	Directly email quotations to customer inboxes via Resend API with custom subject line, personalized body text, and attached official quotation PDF.

■ **DISPATCH TIP:** The integrated email service eliminates the need to download PDFs and open external email clients, logging exact send timestamps for dispatch proof.

MODULE 10: MULTI-USER GOVERNANCE

10. Team Management & Role-Based Access Control (RBAC)



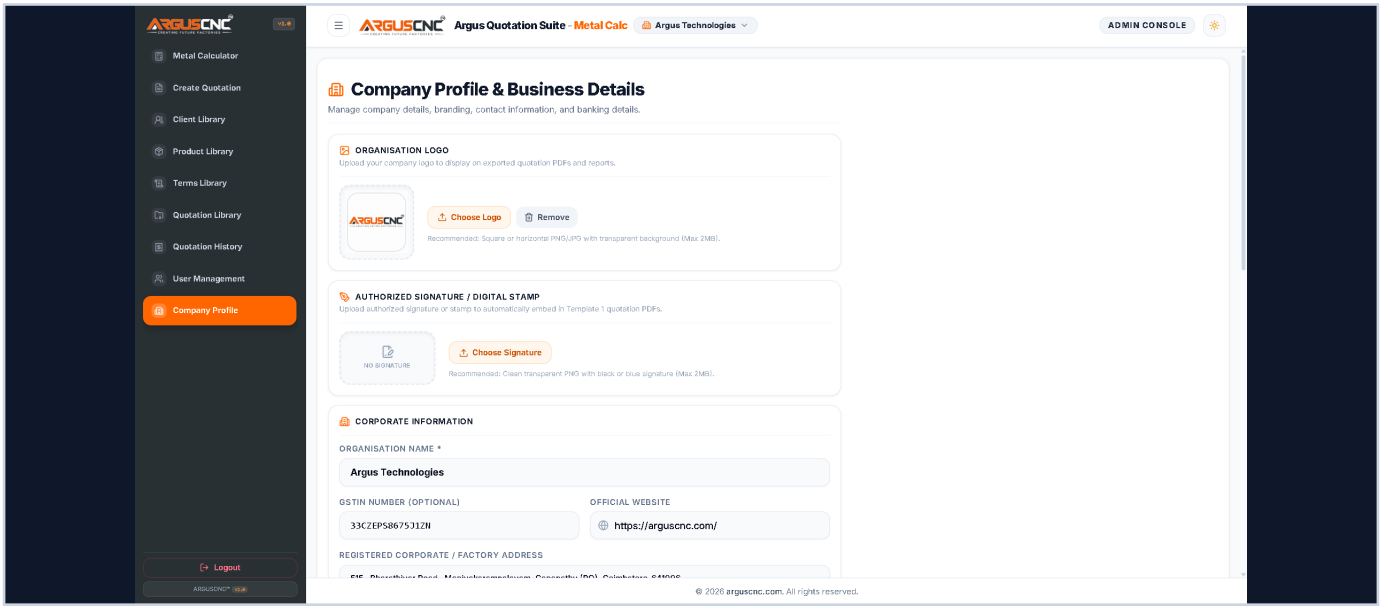
As machine shops grow, multiple estimation engineers, sales representatives, and workshop supervisors collaborate on customer quotes. The Team Directory manages multi-user seats with granular Role-Based Access Controls (RBAC).

Organisation Access Code	Each approved organisation has a secure alphanumeric Access Code (e.g. ARGU-2753). New team members join the organisation instantly using this code.
Granular Permission Toggles	Admins toggle specific capabilities per employee: canAccessCalculator, canAccessQuotation, canViewClients, canEditClients, canAccessProducts, canAccessHistory, canAccessSettings.
Financial Protection	Prevent junior estimators from viewing or altering bottom-line profit percentages, company bank details, or sensitive client master lists.
User Accountability	Every quotation created in the suite logs the specific username (e.g. Created by @priya) for complete traceability and performance tracking.

■ **SECURITY NOTICE:** Restrict 'canAccessSettings' to managing directors to prevent accidental alterations of company bank account details or digital signatures.

MODULE 11: ENTERPRISE CUSTOMIZATION & LEGAL PROFILE

11. Organisation Settings & White-Label Branding



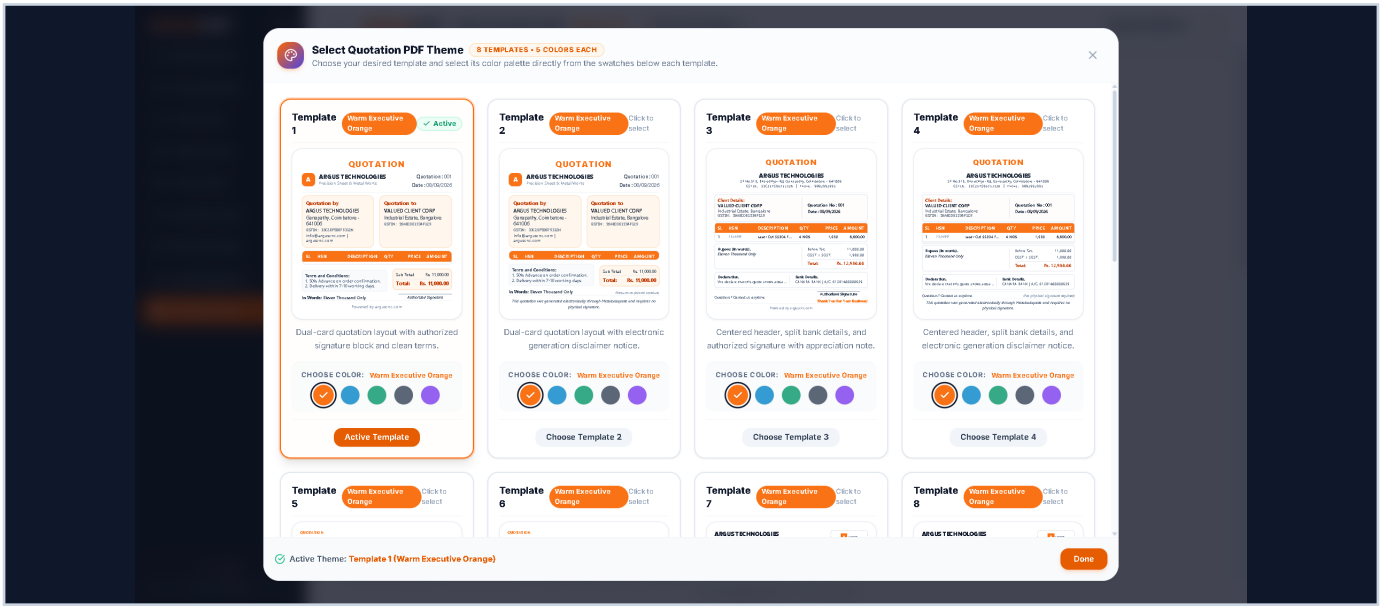
The Organisation Settings module establishes your enterprise's visual identity, legal standing, and banking particulars. It supports multiple sub-companies / sister concerns, high-resolution logo uploads, and authorized signatory digital seals.

Legal Entity Master	Configure Legal Organisation Name, GSTIN (33CZEPS8675J1ZN), PAN number, Official Dispatch Emails, and Primary Registered Works Address.
Brand Assets Upload	Upload high-resolution transparent corporate logos and official authorized signatory digital seals for automatic placement on quotation headers and footers.
Multi-Entity Sister Concerns	Add subsidiary entities or sister machine shops with distinct GSTINs, addresses, and branding, switching between them during quotation drafting.
Banking & RTGS Details	Save Bank Name, Current Account Number, IFSC Code, and Branch information for automated insertion into customer quotation payment instructions.

■ **BRANDING TIP:** Upload your company logo and digital signature in high-resolution PNG format with transparent backgrounds for the crispest PDF rendering.

MODULE 12: DOCUMENT GENERATION & LIVE PREVIEW

12. Professional PDF Quotation Engine & Document Themes



Quotation presentation is decisive in winning high-value manufacturing contracts. The suite features an advanced PDF generation engine offering multiple designer themes, custom color accents, and a real-time Global Live PDF Preview modal.

Multiple Layout Themes	Choose between Modern Blue Corporate, Industrial Blueprint, Classic Elegance, and Clean Minimal layouts tailored for engineering tenders.
Dynamic Brand Accent Colors	Select custom brand accent colors (Argus Orange, Tech Blue, Forest Emerald, Slate Titanium) to match your corporate identity.
Live Global PDF Preview	Inspect the full multi-page quotation document inside the application before printing or emailing, ensuring formatting perfection.
Statutory Tax Breakdown	Complies with Indian GST standards: line-by-line HSN/SAC codes, CGST, SGST, IGST totals, reverse charge declarations, and bank remittance info.

■ **PRESENTATION TIP:** Use the 'Modern Corporate' theme with vibrant accent highlights for commercial tenders, and 'Industrial Blueprint' for detailed technical tooling submissions.

SUMMARY: OPERATIONS & PLATFORM ARCHITECTURE

System Specifications & Best Practices Summary

Summary of Platform Capabilities:
The **Argus Quotation Suite & Metal Calculator** delivers an end-to-end digital workflow that bridges the gap between workshop CAD engineering and front-office commercial sales. By integrating geometric CAD volume calculations, daily live commodity market rates, AI-driven cycle time estimation, multi-entity branding, and seamless PDF quotation generation, the suite empowers modern CNC job shops and precision manufacturers to win more tenders, eliminate estimation errors, and accelerate cash flow.

Technical Specification	Enterprise Configuration & Performance Benchmark
Application Architecture	Full-Stack Node.js Express Backend, Vanilla JS Responsive Single-Page Application, PyWebView Desktop Wra
Database Layer	MongoDB Atlas Enterprise Cloud Cluster with automatic failover, cold-start pre-warming, and local caching
AI Multimodal Model	Google Gemini AI Vision API for 2D engineering drawing cycle time and tooling parameter extraction
Commodity Pricing API	Metals-API Daily Real-Time Commodity Feed for Carbon Steel, Stainless Steel, Aluminium, Copper, Brass
GST Integration	Appyflow Indian Goods & Services Tax (GST) Verification API for automated trade name and address resolution
Email Delivery Service	Resend Transactional Email API for direct PDF attachment dispatch with deliverability logging
Desktop Runtime	Standalone single-file executable (MetalCalculator.exe) bundling Node 20 runtime and EdgeChromium WebVie
Supported Platforms	Windows 10/11 Desktop, macOS, Linux, and Modern Web Browsers (Chrome, Edge, Safari, Firefox)

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Precision CNC Machining • VMC Tooling • Advanced Manufacturing Software Solutions